

HiSUP 2.0 - University ERP System

Ready-to-Build Implementation Plan

Project Vision

HiSUP 2.0 is a database-driven University ERP platform. The system is centered on business workflows rather than simple CRUD operations. Every module must interact through defined relationships, transactions, audit logs, notifications, and role-based permissions.

Architecture

Frontend: React + Vite + Tailwind CSS. Backend: Node.js + Express + JWT + Supabase SDK. Database: Supabase PostgreSQL. Frontend communicates only with backend APIs. Backend enforces business rules and accesses PostgreSQL.

User Roles

Admin, Student, Faculty, and Finance Officer. Each role receives only the permissions required for its workflows.

Database Core Modules

Authentication, Academic Management, Enrollment, Attendance, Results, Finance, Library, Hostel, Notifications, Audit Logs, Announcements, Academic Calendar, Support Tickets, Transcript Requests, Course Materials, and Exam Scheduling.

Key Database Tables

user_accounts, departments, programs, semesters, academic_sessions, students, faculty, courses, course_prerequisites, sections, enrollments, attendance_records, grades, results, fee_structures, student_fees, fee_payments, receipts, library_items, library_issues, library_fines, hostels, rooms, hostel_allotments, notifications, audit_logs, announcements, support_tickets, transcript_requests, course_materials, exam_schedule.

Business Workflow: Enrollment

Validate student status, fee clearance, prerequisites, duplicate enrollment, and seat availability. Execute transaction. Create enrollment, update seats, create notification, create audit log.

Business Workflow: Finance

Student uploads payment proof. Finance verifies payment. System updates balances, generates receipt, sends notification, and writes audit logs.

Business Workflow: Results

Faculty uploads grades. Trigger recalculates GPA and CGPA. Results become available to students automatically.

Stored Procedures

RegisterStudent, EnrollStudentInCourse, AssignFacultyToSection, UploadGrades, GenerateResult, ApprovePayment, GenerateFeeChallan, IssueBook, ReturnBook, AllocateRoom, GenerateTranscript.

Functions

fn_calculate_cgpa, fn_attendance_percentage, fn_outstanding_fee, fn_library_fine, fn_available_seats, fn_student_status.

Triggers

Enrollment trigger, grade trigger, payment trigger, library return trigger, hostel allocation trigger, universal audit trigger.

Views

vw_student_dashboard, vw_admin_dashboard, vw_faculty_dashboard, vw_finance_dashboard, vw_fee_defaulters, vw_library_status, vw_course_enrollment_stats, vw_hostel_occupancy.

Frontend Design

Professional academic interface. Minimal radius (6px), subtle hover effects, sidebar layout, dashboard cards, charts, tables, and forms. Use real campus imagery instead of generic icons.

Homepage Structure

Hero section with campus image, feature showcase cards, statistics section, academic calendar timeline, announcements section, and login access.

Student Dashboard

Profile summary, CGPA, attendance, enrolled courses, fee status, library status, notifications, GPA trends, and attendance charts.

Faculty Dashboard

Assigned courses, attendance progress, grade submission status, teaching workload, and course materials.

Finance Dashboard

Revenue summaries, pending approvals, fee defaulters, collection trends, and receipt management.

Admin Dashboard

University statistics, audit logs, announcements, academic calendar management, user management, and analytics.

Development Roadmap

Phase 1: Database schema and ERD. Phase 2: Authentication. Phase 3: Academic modules.
Phase 4: Enrollment and attendance. Phase 5: Results and finance. Phase 6: Library and hostel.
Phase 7: Reports, dashboards, testing, deployment, and documentation.

Viva Focus

Explain relationships, foreign keys, normalization, transactions, triggers, procedures, functions, views, indexes, and workflow logic.